Pearson Edexcel IGCSE

OL Modular Mathematics (4MA1)

Monthly Exam 1 (OL 2027) — Complete Model Answers

Exam Details: Total Marks: **75** | Time Allowed: **90 Minutes** | Questions: **26**
Purpose: Full worked solutions with step-by-step methodology (**[M1]** Method, **[A1]** Accuracy marks) designed for teacher correction and student feedback.

SECTION A: Foundational Algebra Skills (Questions 1–5)

Question 1 **(2 Marks)**

Expand and simplify completely: $3(2x-5)-2(x-4)$

Detailed Step-by-Step Solution:

1. **Step 1 (Expand brackets):** Multiply 3 into the first bracket and -2 into the second bracket:
$$3(2x-5)-2(x-4)=6x-15-2x+8 \quad \textbf{[M1]}$$
2. **Step 2 (Collect like terms):** Group the x-terms and constant terms:
$$(6x-2x)+(-15+8)=4x-7 \quad \textbf{[A1]}$$

Final Answer

Answer: $4x-7$

Question 2 **(2 Marks)**

Factorise completely: $15ab^2-10a^2b$

Detailed Step-by-Step Solution:

1. **Step 1 (Identify Highest Common Factor):** The HCF of numbers 15 and 10 is 5. The HCF of variables ab^2 and a^2b is ab. Thus, HCF $=5ab$. **[M1]**
2. **Step 2 (Factor out $5ab$):** Divide each term by $5ab$:
$$15ab^2-10a^2b=5ab(3b-2a) \quad \textbf{[A1]}$$

Final Answer

Answer: $5ab(3b-2a)$

Question 3 **(2 Marks)**

Solve the linear equation: $\dfrac{3x+2}{4}=5$

Detailed Step-by-Step Solution:

1. **Step 1 (Clear the denominator):** Multiply both sides of the equation by 4:

$$3x + 2 = 5 \times 4 \implies 3x + 2 = 20 \quad \textbf{[M1]}$$

2. **Step 2 (Isolate x):** Subtract 2 from both sides and divide by 3:

$$3x = 18 \implies x = \frac{18}{3} = 6 \quad \textbf{[A1]}$$

Final Answer

Answer: $x = 6$

Question 4 **(2 Marks)**

Solve the linear inequality: $5x - 7 \leq 3x + 9$

Detailed Step-by-Step Solution:

1. **Step 1 (Collect x terms on LHS):** Subtract $3x$ from both sides:

$$5x - 3x - 7 \leq 9 \implies 2x - 7 \leq 9 \quad \textbf{[M1]}$$

2. **Step 2 (Isolate x):** Add 7 to both sides and divide by 2:

$$2x \leq 16 \implies x \leq 8 \quad \textbf{[A1]}$$

Final Answer

Answer: $x \leq 8$

Question 5 **(2 Marks)**

Make t the subject of the formula: $v = u + at$

Detailed Step-by-Step Solution:

1. **Step 1 (Isolate at term):** Subtract u from both sides:

$$v - u = at \implies at = v - u \quad \textbf{[M1]}$$

2. **Step 2 (Divide by a):**

$$t = \frac{v - u}{a} \quad \textbf{[A1]}$$

Final Answer

Answer: $t = \dfrac{v - u}{a}$

SECTION B: Core Higher Tier Skills (Questions 6–12)

Question 6 **(2 Marks)**

Expand and simplify completely: $(2x+3)(x-5)$

Detailed Step-by-Step Solution:

1. **Step 1 (Expand using FOIL):**

$$(2x+3)(x-5) = 2x(x) + 2x(-5) + 3(x) + 3(-5) = 2x^2 - 10x + 3x - 15 \quad \textbf{[M1]}$$

2. **Step 2 (Combine like terms):**

$$2x^2 - 7x - 15 \quad \textbf{[A1]}$$

Final Answer

Answer: $2x^2 - 7x - 15$

Question 7 **(2 Marks)**

Factorise completely: $x^2 - 9x + 20$

Detailed Step-by-Step Solution:

1. **Step 1 (Find factor pairs of $+20$ that sum to -9):** The pair is -4 and -5 since $(-4)\times(-5) = 20$ and $(-4)+(-5) = -9$. **[M1]**
2. **Step 2 (Write in factored form):**

$$(x-4)(x-5) \quad \textbf{[A1]}$$

Final Answer

Answer: $(x-4)(x-5)$

Question 8 **(3 Marks)**

Express as a single fraction in its simplest form: $\dfrac{2}{x+1} + \dfrac{3}{x-2}$

Detailed Step-by-Step Solution:

1. **Step 1 (Find common denominator $(x+1)(x-2)$):**

$$\frac{2(x-2) + 3(x+1)}{(x+1)(x-2)} \quad \textbf{[M1]}$$

2. **Step 2 (Expand and simplify numerator):**

$$2(x-2) + 3(x+1) = 2x - 4 + 3x + 3 = 5x - 1 \quad \textbf{[M1]}$$

3. **Step 3 (Write single fraction):**

$$\frac{5x-1}{(x+1)(x-2)} \quad \textbf{[A1]}$$

Final Answer

Answer: $\dfrac{5x-1}{(x+1)(x-2)}$

Question 9 **(3 Marks)**

Solve the quadratic equation by factorisation: $x^2+2x-15=0$

Detailed Step-by-Step Solution:

1. **Step 1 (Factorise quadratic):** Find factors of -15 that sum to $+2$: $+5$ and -3.

$$(x+5)(x-3)=0 \quad \textbf{[M1, M1]}$$

2. **Step 2 (Set factors to zero and solve):**

$$x+5=0 \implies x=-5 \quad \text{or} \quad x-3=0 \implies x=3 \quad \textbf{[A1]}$$

Final Answer

Answer: $x=-5$ or $x=3$

Question 10 **(3 Marks)**

Solve the compound linear inequality: $-5<3x+1\leq 13$

Detailed Step-by-Step Solution:

1. **Step 1 (Subtract 1 from all parts):**

$$-5-1<3x\leq 13-1 \implies -6<3x\leq 12 \quad \textbf{[M1]}$$

2. **Step 2 (Divide all parts by 3):**

$$\frac{-6}{3}<\frac{3x}{3}\leq\frac{12}{3} \implies -2<x\leq 4 \quad \textbf{[M1, A1]}$$

Final Answer

Answer: $-2<x\leq 4$

Question 11 **(3 Marks)**

Rearrange the formula to make x the subject: $y = \dfrac{2x+5}{x-3}$

Detailed Step-by-Step Solution:

1. **Step 1 (Multiply by denominator $(x-3)$):**

$$y(x-3) = 2x+5 \implies yx - 3y = 2x+5 \quad \textbf{[M1]}$$

2. **Step 2 (Collect x terms on LHS and group):**

$$yx - 2x = 3y+5 \implies x(y-2) = 3y+5 \quad \textbf{[M1]}$$

3. **Step 3 (Divide by $(y-2)$):**

$$x = \frac{3y+5}{y-2} \quad \textbf{[A1]}$$

Final Answer

Answer: $x = \dfrac{3y+5}{y-2}$

Question 12 **(3 Marks)**

Solve the linear simultaneous equations: $3x + 2y = 16$ and $2x - 3y = 2$

Detailed Step-by-Step Solution:

1. **Step 1 (Scale equations to eliminate y):** Multiply Eq(1) by 3 and Eq(2) by 2:

$$9x + 6y = 48 \quad \text{and} \quad 4x - 6y = 4 \quad \textbf{[M1]}$$

2. **Step 2 (Add scaled equations):**

$$(9x + 4x) = 48 + 4 \implies 13x = 52 \implies x = 4 \quad \textbf{[M1]}$$

3. **Step 3 (Substitute $x = 4$ to find y):**

$$3(4) + 2y = 16 \implies 12 + 2y = 16 \implies 2y = 4 \implies y = 2 \quad \textbf{[A1]}$$

Final Answer

Answer: $x = 4, \quad y = 2$

SECTION C: Advanced Higher Mastery (Questions 13–18)

Question 13 **(3 Marks)**

Expand and simplify completely: $(3x-2y)^2-(2x+y)(x-3y)$

Detailed Step-by-Step Solution:

1. **Step 1 (Expand $(3x-2y)^2$):**

$$(3x-2y)^2=9x^2-12xy+4y^2 \quad \textbf{[M1]}$$

2. **Step 2 (Expand $(2x+y)(x-3y)$):**

$$(2x+y)(x-3y)=2x^2-6xy+xy-3y^2=2x^2-5xy-3y^2 \quad \textbf{[M1]}$$

3. **Step 3 (Subtract second expansion from first):**

$$(9x^2-12xy+4y^2)-(2x^2-5xy-3y^2)=9x^2-12xy+4y^2-2x^2+5xy+3y^2$$

$$=7x^2-7xy+7y^2 \quad \textbf{[A1]}$$

Final Answer

Answer: $7x^2-7xy+7y^2$

Question 14 **(3 Marks)**

Factorise completely: $6x^2-11x-10$

Detailed Step-by-Step Solution:

1. **Step 1 (Find factors of $6\times(-10)=-60$ that sum to -11):** The pair is -15 and $+4$. **[M1]**
2. **Step 2 (Split middle term and group):**

$$6x^2-15x+4x-10=3x(2x-5)+2(2x-5) \quad \textbf{[M1]}$$

3. **Step 3 (Factor out $(2x-5)$):**

$$(2x-5)(3x+2) \quad \textbf{[A1]}$$

Final Answer

Answer: $(2x-5)(3x+2)$

Question 15 **(3 Marks)**

Simplify the algebraic fraction completely: $\dfrac{2x^2-8}{x^2+5x+6}$

Detailed Step-by-Step Solution:

1. **Step 1 (Factorise numerator):** Take out 2 then use difference of two squares:

$$2(x^2-4)=2(x-2)(x+2) \quad \textbf{[M1]}$$

2. **Step 2 (Factorise denominator):**

$$x^2 + 5x + 6 = (x+2)(x+3) \quad \textbf{[M1]}$$

3. **Step 3 (Cancel common factor $(x+2)$):**

$$\frac{2(x-2)(x+2)}{(x+2)(x+3)} = \frac{2(x-2)}{x+3} \quad \textbf{[A1]}$$

Final Answer

Answer: $\dfrac{2(x-2)}{x+3}$

Question 16 **(3 Marks)**

Solve the algebraic fraction equation: $\dfrac{4}{x-2} - \dfrac{3}{x+1} = 1$

Detailed Step-by-Step Solution:

1. **Step 1 (Clear denominators by multiplying by $(x-2)(x+1)$):**

$$4(x+1) - 3(x-2) = 1(x-2)(x+1) \quad \textbf{[M1]}$$

2. **Step 2 (Expand and form quadratic equation):**

$$4x + 4 - 3x + 6 = x^2 - x - 2 \implies x + 10 = x^2 - x - 2 \implies x^2 - 2x - 12 = 0 \quad \textbf{[M1]}$$

3. **Step 3 (Factorise and solve):**

$$(x-4)(x+3) = 0 \implies x = 4 \quad \text{or} \quad x = -3 \quad \textbf{[A1]}$$

Final Answer

Answer: $x = 4$ or $x = -3$

Question 17 **(3 Marks)**

Solve the quadratic inequality: $x^2 - 3x - 10 > 0$

Detailed Step-by-Step Solution:

1. **Step 1 (Find critical values by setting expression to 0):**

$$x^2 - 3x - 10 = 0 \implies (x-5)(x+2) = 0 \implies x = 5 \quad \text{and} \quad x = -2 \quad \textbf{[M1]}$$

2. **Step 2 (Determine valid region for > 0):** Since coefficient of x^2 is positive, the curve opens upwards. Positive region is strictly outside the roots. **[M1]**

3. **Step 3 (Write solution):**

$$x < -2 \quad \text{or} \quad x > 5 \quad \textbf{[A1]}$$

Final Answer

Answer: $x < -2 \quad \text{or} \quad x > 5$

Question 18 **(3 Marks)**

Make x the subject of the formula: $r = \sqrt{\dfrac{3x+1}{x-2}}$

Detailed Step-by-Step Solution:

1. **Step 1 (Square both sides to remove square root):**

$$r^2 = \frac{3x+1}{x-2} \implies r^2(x-2) = 3x+1 \quad \textbf{[M1]}$$

2. **Step 2 (Expand and group x terms on LHS):**

$$r^2x - 2r^2 = 3x + 1 \implies r^2x - 3x = 2r^2 + 1 \implies x(r^2-3) = 2r^2+1 \quad \textbf{[M1]}$$

3. **Step 3 (Isolate x):**

$$x = \frac{2r^2+1}{r^2-3} \quad \textbf{[A1]}$$

Final Answer

Answer: $x = \dfrac{2r^2+1}{r^2-3}$

SECTION D: Extension Problem Solving (Questions 19–22)

Question 19 **(3 Marks)**

Expand and simplify the triple binomial product completely: $(2x+1)(x-3)(3x+2)$

Detailed Step-by-Step Solution:

1. **Step 1 (Expand first pair of brackets):**

$$(2x+1)(x-3) = 2x^2 - 6x + x - 3 = 2x^2 - 5x - 3 \quad \textbf{[M1]}$$

2. **Step 2 (Multiply result by third bracket $(3x+2)$):**

$$(2x^2 - 5x - 3)(3x+2) = 2x^2(3x+2) - 5x(3x+2) - 3(3x+2) \quad \textbf{[M1]}$$

$$= 6x^3 + 4x^2 - 15x^2 - 10x - 9x - 6$$

3. **Step 3 (Combine like terms):**

$$6x^3 - 11x^2 - 19x - 6 \quad \textbf{[A1]}$$

Final Answer

Answer: $6x^3 - 11x^2 - 19x - 6$

Question 20 **(3 Marks)**

Factorise the expression completely by substitution or expansion: $2(x-3)^2 - 5(x-3) - 3$

Detailed Step-by-Step Solution:

1. **Step 1 (Substitute $u = x - 3$):**

$$2u^2 - 5u - 3 \quad \textbf{[M1]}$$

2. **Step 2 (Factorise quadratic in u):**

$$(2u+1)(u-3) \quad \textbf{[M1]}$$

3. **Step 3 (Substitute back $u = x - 3$ and simplify):**

$$[2(x-3)+1][(x-3)-3] = [2x-6+1][x-6] = (2x-5)(x-6) \quad \textbf{[A1]}$$

Final Answer

Answer: $(2x-5)(x-6)$

Question 21 **(3 Marks)**

Solve the non-unit quadratic inequality: $3x^2 - 10x - 8 \leq 0$

Detailed Step-by-Step Solution:

1. **Step 1 (Find critical values by setting quadratic to 0):**

$$3x^2 - 10x - 8 = 0 \implies (3x+2)(x-4) = 0 \implies x = -\frac{2}{3} \quad \text{and} \quad x = 4 \quad \textbf{[M1, M1]}$$

2. **Step 2 (Determine enclosed region for ≤ 0):**

$$-\frac{2}{3} \leq x \leq 4 \quad \textbf{[A1]}$$

Final Answer

Answer: $-\frac{2}{3} \leq x \leq 4$

Question 22 **(4 Marks)**

Solve the non-linear simultaneous equations: $y = 2x - 3$ and $x^2 + y^2 = 18$

Detailed Step-by-Step Solution:

1. **Step 1 (Substitute linear equation into circle equation):**

$$x^2 + (2x - 3)^2 = 18 \quad \textbf{[M1]}$$

2. **Step 2 (Expand and form standard quadratic equation):**

$$x^2 + 4x^2 - 12x + 9 = 18 \implies 5x^2 - 12x - 9 = 0 \quad \textbf{[M1]}$$

3. **Step 3 (Solve quadratic for x):**

$$(5x + 3)(x - 3) = 0 \implies x = 3 \quad \text{or} \quad x = -0.6 \left(-\frac{3}{5}\right) \quad \textbf{[A1]}$$

4. **Step 4 (Calculate corresponding y values using $y = 2x - 3$):**

- When $x = 3 \implies y = 2(3) - 3 = 3$
- When $x = -0.6 \implies y = 2(-0.6) - 3 = -1.2 - 3 = -4.2$

[A1]

Final Answer

Answer: $(x = 3, y = 3)$ and $(x = -0.6, y = -4.2)$

SECTION E: Advanced Algebra Challenge (Questions 23–26)

Question 23 (4 Marks)

Show that the expression simplifies to 1 for all valid values of x:

$$\frac{x^2-9}{2x^2+5x-3} \div \frac{x^2-3x}{2x^2-x} = 1$$

Detailed Step-by-Step Solution:

1. **Step 1 (Factorise all four quadratic/algebraic terms):**
   - $x^2-9 = (x-3)(x+3)$
   - $2x^2+5x-3 = (2x-1)(x+3)$
   - $x^2-3x = x(x-3)$
   - $2x^2-x = x(2x-1)$

   [M1, M1]

2. **Step 2 (Invert second fraction and multiply):**

$$\frac{(x-3)(x+3)}{(2x-1)(x+3)} \times \frac{x(2x-1)}{x(x-3)} \quad \textbf{[M1]}$$

3. **Step 3 (Cancel all common factors in numerator and denominator):**

$$= \frac{(x-3)(x+3)}{(2x-1)(x+3)} \times \frac{x(2x-1)}{x(x-3)} = 1 \quad \textbf{[A1]}$$

Final Answer

Proof Complete: Expression simplifies exactly to 1.

Question 24 (4 Marks)

Given the formula: $y = \dfrac{x^2+11}{2x^2-3}$

(a) Make x the subject of the formula.

(b) Hence, find the exact positive value of x when $y = 3$.

Detailed Step-by-Step Solution:

1. **Part (a) – Rearrange for x:**
   - Multiply by denominator: $y(2x^2-3) = x^2+11 \implies 2yx^2-3y = x^2+11$ **[M1]**
   - Collect x^2 terms: $2yx^2-x^2 = 3y+11 \implies x^2(2y-1) = 3y+11$
   - Divide and take square root: $x^2 = \frac{3y+11}{2y-1} \implies x = \sqrt{\frac{3y+11}{2y-1}}$ **[M1, A1]**

2. **Part (b) – Substitute $y = 3$:**

$$x = \sqrt{\frac{3(3)+11}{2(3)-1}} = \sqrt{\frac{9+11}{6-1}} = \sqrt{\frac{20}{5}} = \sqrt{4} = 2 \quad \textbf{[A1]}$$

Final Answer

(a) $x = \sqrt{\dfrac{3y+11}{2y-1}}$ **(b)** $x = 2$

Question 25 (4 Marks)

Solve the simultaneous quadratic inequality system: $2x^2 - 7x - 15 \leq 0$ AND $x^2 - 4x + 3 > 0$. State the complete set of values of x that satisfy both inequalities.

Detailed Step-by-Step Solution:

1. **Step 1 (Solve $2x^2 - 7x - 15 \leq 0$):**

$$(2x+3)(x-5) \leq 0 \implies \text{Critical values } x = -\frac{3}{2}, 5 \implies -\frac{3}{2} \leq x \leq 5 \quad \textbf{[M1, A1]}$$

2. **Step 2 (Solve $x^2 - 4x + 3 > 0$):**

$$(x-1)(x-3) > 0 \implies \text{Critical values } x = 1, 3 \implies x < 1 \quad \text{or} \quad x > 3 \quad \textbf{[M1]}$$

3. **Step 3 (Find intersection of both inequalities):**

- Lower region: $[-\frac{3}{2}, 5] \cap (-\infty, 1) \implies -\frac{3}{2} \leq x < 1$
- Upper region: $[-\frac{3}{2}, 5] \cap (3, \infty) \implies 3 < x \leq 5$

[A1]

Final Answer

Answer: $-\frac{3}{2} \leq x < 1 \quad \text{or} \quad 3 < x \leq 5$

Question 26 (5 Marks)

Solve the non-linear simultaneous equations containing product terms:

$$2x - y = 5 \quad \text{and} \quad 2x^2 - xy - y^2 = 9$$

Detailed Step-by-Step Solution:

1. **Step 1 (Rearrange linear equation for y):**

$$y = 2x - 5 \quad \textbf{[M1]}$$

2. **Step 2 (Substitute into quadratic equation):**

$$2x^2 - x(2x-5) - (2x-5)^2 = 9 \quad \textbf{[M1]}$$

3. **Step 3 (Expand and collect like terms):**

$$2x^2 - (2x^2 - 5x) - (4x^2 - 20x + 25) = 9$$

$$2x^2 - 2x^2 + 5x - 4x^2 + 20x - 25 = 9 \implies -4x^2 + 25x - 34 = 0$$

$$4x^2 - 25x + 34 = 0 \quad \textbf{[M1]}$$

4. **Step 4 (Solve quadratic for x):**

$$(4x - 17)(x - 2) = 0 \implies x = 2 \quad \text{or} \quad x = \frac{17}{4} = 4.25 \quad \textbf{[A1]}$$

5. **Step 5 (Find corresponding y values using $y = 2x - 5$):**

- When $x = 2 \implies y = 2(2) - 5 = -1$
- When $x = 4.25 \implies y = 2(4.25) - 5 = 8.5 - 5 = 3.5$

[A1]

Final Answer

Answer: $(x = 2, y = -1)$ and $(x = 4.25, y = 3.5)$